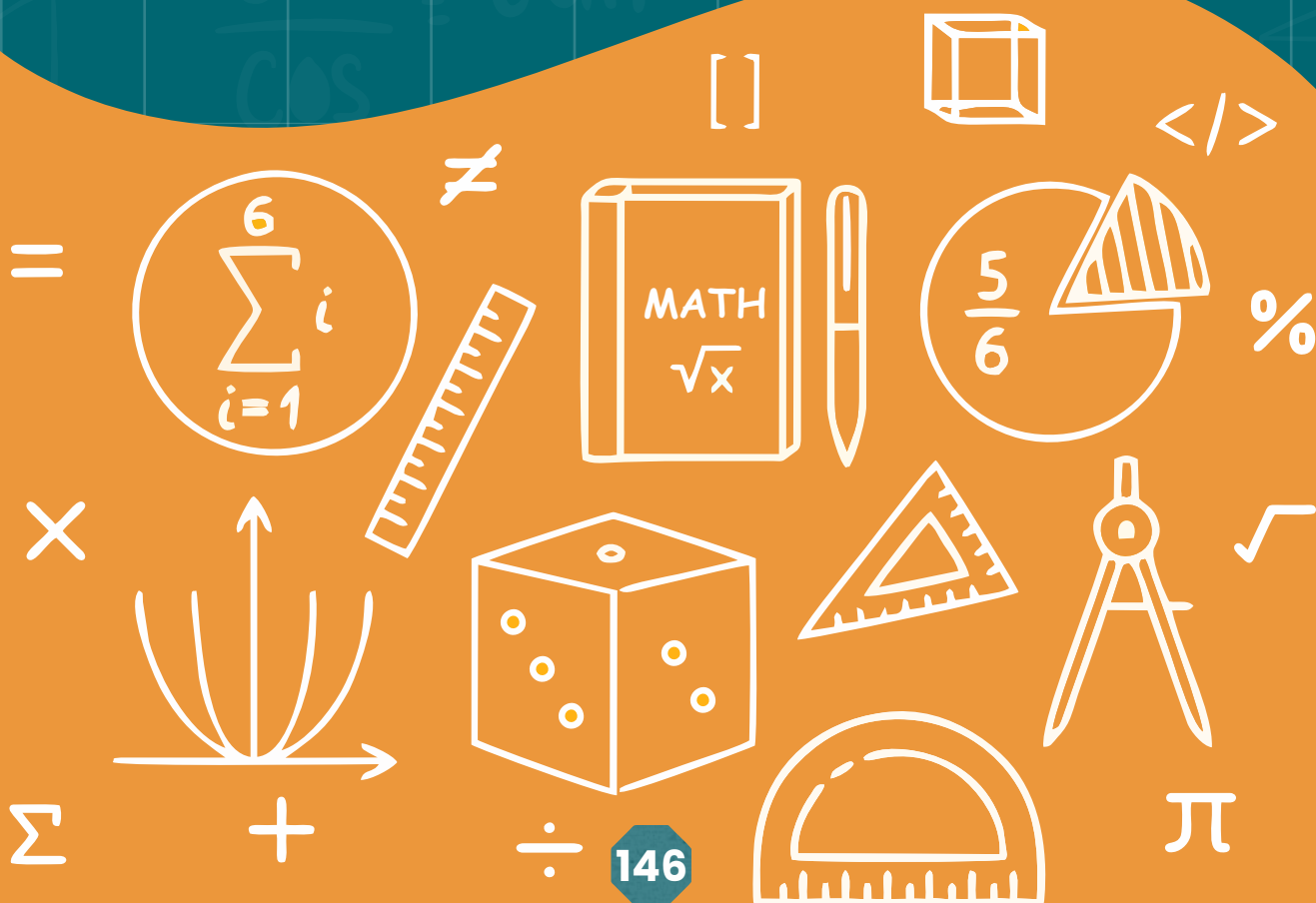


UNIT 13

INFORMATION HANDLING



FREQUENCY DISTRIBUTION

Frequency and Tally Charts

Frequency refers to the number of times a value occurs in a group of data. A tally chart is a simple way to organize and record data using groups of five. The symbol 'I' is used to represent one, and every fifth mark is crossed to form a group of five (e.g., ) . Tally marks are quick and efficient for counting and organizing data



- Tally charts can be used for grouped and ungrouped data:
- Ungrouped data is data given as separate pieces. It is useful when there is a small number of outcomes.
- Grouped data is used when there are large numbers of possible outcomes. It makes the data easier to read.

Example:

Let's consider the marks of 25 students in standard IX in a particular school. The scores are as follows:

41, 53, 64, 31, 53, 33, 70, 61, 74, 32, 53, 56, 56, 64, 56, 88, 28, 70, 56, 64, 74, 53, 53, 61, 31.

The maximum mark allotted to the paper is 100. Draw a tally chart for the above data.

Solution: Rearranged data in ascending order:

Score	Tally marks	Frequency
28		1
31		2
32		1
33		1

41		1
53		5
56		4
61		2
64		3
70		2
74		2
88		1
Total		25

EXERCISE 13.1

1. Following are the scores of students of class 3 in a maths test:

12, 12, 14, 15, 16, 18, 13, 15, 16, 17, 18, 13, 16, 20, 20, 19, 17, 15, 12, 15, 16, 17, 18, 18, 19, 20, 16, 19.

Make a tally marks table.

2. Draw a tally marks table for the following data:

A, B, A, B, C, D, E, D, C, B, A, E, C, D, B, E, A, E, C, D, B, E, C.

3. There are 30 students in class VII who secured following marks in a Mathematics Test:

45, 19, 49, 17, 37, 10, 25, 28, 50, 46, 43, 34, 25, 50, 48, 46, 42,
34, 33, 11, 28, 19, 30, 17, 13, 11, 39, 41, 35, 34

Make a frequency distribution table using 5 as a class interval of an equal size.

PIE GRAPH

A pie chart is a circular representation of data, where the circle is divided into slices (sectors) that look like pieces of a pie. Each slice's size represents the proportion of a particular category in the data. The area of each slice is directly proportional to its frequency, and the central angle of each sector is calculated using the formula: $(\text{Given Data} / \text{Total Data}) \times 360^\circ$.

Learn about [how to draw a pie chart](#).

Some important Formulas:

- To draw a pie chart with percentages

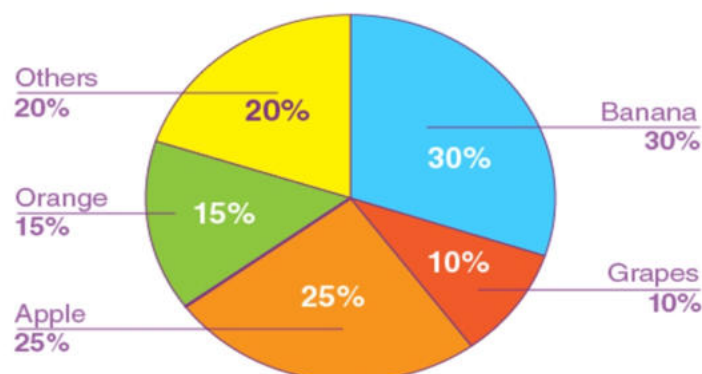
Percentage of a data = $(\text{Given data} / \text{Total of all data}) \times 100$

- To draw a pie chart with central angle

Central angle of a data = $(\text{Given data} / \text{Total of all data}) \times 360^\circ$

Example:

The below pie chart shows the sale of different fruits in a day for a shop:



Answer the following questions based on the pie chart:

- (i) If a total of 1200 kg of fruits were sold in a day, calculate the amount of bananas sold (in kg).
 (ii) Find the difference between sales of grapes and oranges.
 (iii) Calculate the central angle for 'others'.

Solution:

(i) Total amount of fruits sold = 1200 kg

Percentage of banana sold = 30%

Amount of banana sold = $1200 \times (30/100) = 360$ kg

(ii) Amount of grapes sold = $1200 \times (10/100) = 120$ kg

Amount of oranges sold = $1200 \times (15/100) = 180$ kg

Difference = $180 - 120 = 60$ kg

(iii) Amount of other fruits sold = $1200 \times 20 = 240$ kg

Central angle = $240/1200 \times 360^\circ = 72^\circ$.

Example:

A person spends his time on different activities daily (in hours):

Activity	Office Work	Exercise	Travelling	Watching shows	Sleeping	Misc.
Number of hours	9	1	2	3	7	2

Draw a pie chart for this information.

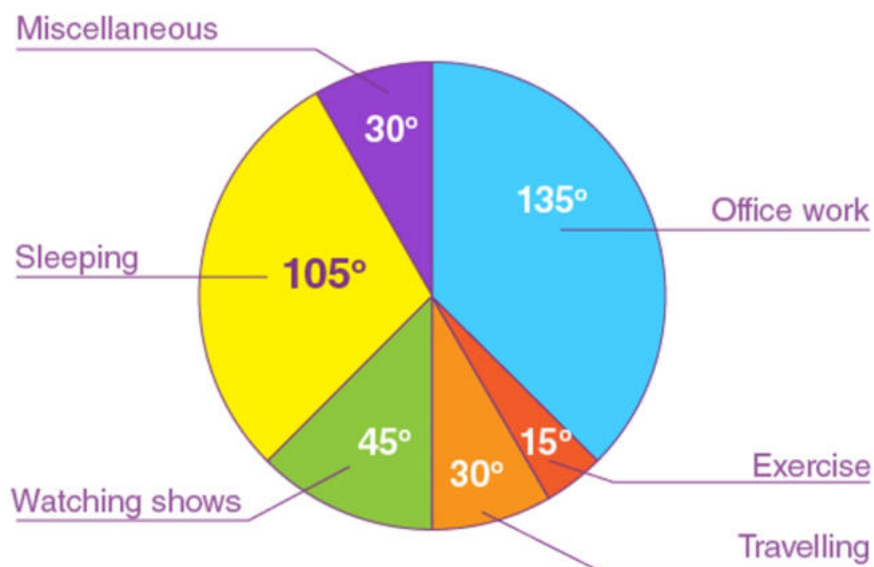
Solution:

First, we have to calculate the central angles.

Activity	Number of hours	Central angle
Office work	9	$9/24 \times 360^\circ = 135^\circ$
Exercise	1	$1/24 \times 360^\circ = 15^\circ$
Travelling	2	$2/24 \times 360^\circ = 30^\circ$

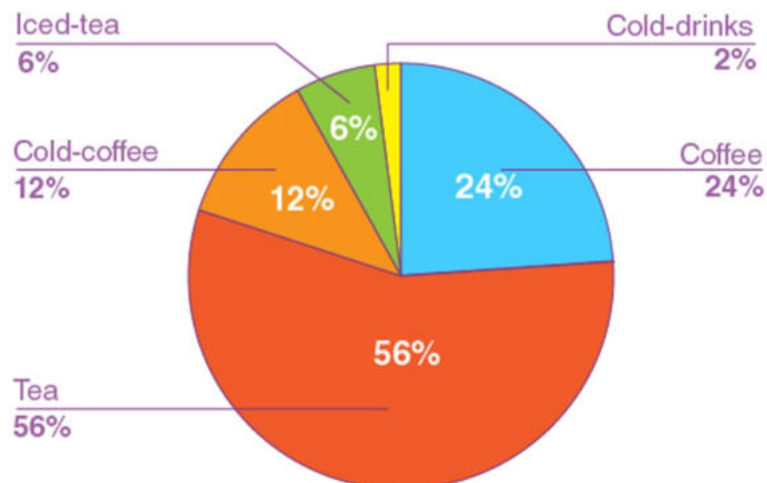
Watching shows	3	$3/24 \times 360^\circ = 45^\circ$
Sleeping	7	$7/24 \times 360^\circ = 105^\circ$
Miscellaneous	2	$2/24 \times 360^\circ = 30^\circ$
Total	24	360°

The pie chart is –



EXERCISES 13.2

Q1. In the summer, a survey was conducted among 400 people about their favourite beverages. The following pie chart shows the data:



Answer the following questions:

- (i) How many people like tea?
- (ii) How many more people like coffee than cold coffee?
- (iii) What is the total central angle for iced tea and cold-drinks?

Q2. There are 60 students in a class, following table shows the data of their result.

Result	First Class	Second Class	Third Class	Fail
Number of students	35%	45%	15%	5%

Draw a pie chart for this information.

Q3. The following data shows the distance in km that Hooria travelled in last month.

90, 44, 55, 49, 28, 9, 92, 27, 18, 84, 50, 60, 79, 69, 24, 89, 63, 74, 35, 48, 39, 80.

Construct the frequency table using 6 as class interval of an equal size. Also draw its pie chart.